

Prince Gabrielle Jhon M. Libertad

princegabriellejhonl@gmail.com • +63 960 346 9262 • +63 926 869 6011
[LinkedIn](#) • github.com/ASTRALLIBERTAD • Manila, Philippines

EDUCATION

Polytechnic University of the Philippines (PUP)

Bachelor of Science in Computer Engineering | Manila, Philippines

2025 – Present

PROJECTS

MicroShell-ID — Marine Microgastropod Identification Tool

In Progress (Private)

- Flutter desktop frontend + AmScope MU1000 USB microscope integration via UVC APIs
- Rust backend with ONNX inference model; hybrid AI + morphological character decision engine
- Flutter–Rust FFI bridge; supports fully offline deployment

LMS Alternative — Learning Management System

- Python app with Google Drive content storage and Firebase authentication
- [Repo: github.com/ASTRALLIBERTAD/LMS-alternative](https://github.com/ASTRALLIBERTAD/LMS-alternative)

NH3 Monitoring System — BLE Ammonia Sensor

- ESP32 BLE sensor with native Kotlin Android app for real-time ammonia monitoring
- [Repo: github.com/ASTRALLIBERTAD/NH3-Monitoring-System](https://github.com/ASTRALLIBERTAD/NH3-Monitoring-System)

Brain — Custom Compiled Programming Language

In Progress

- Compiler written in Rust targeting LLVM IR; lexer, parser, and IR emission in progress
- [Repo: github.com/ASTRALLIBERTAD/Brain](https://github.com/ASTRALLIBERTAD/Brain)

CERTIFICATIONS & TRAINING

Huawei ICT Academy

- Python Programming Basics — January 2026 (Code: EBG20260108002690)
- Algorithm and Program Design — September 2025 (Code: ICT202509190000003)

TECHNICAL SKILLS

Languages: Rust, Python, Kotlin, Dart (Flutter), C++

Frameworks & Libraries: Flutter, TensorFlow, ONNX Runtime, Firebase, Android SDK

Tools & Platforms: Git, CMake, Ninja, LLVM IR, Google Drive API, BLE / UVC protocols

Domains: Computer Vision, IoT & Embedded Systems, Compiler Design, Mobile Development